卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-SG-0202 30-B003701S-S0202

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

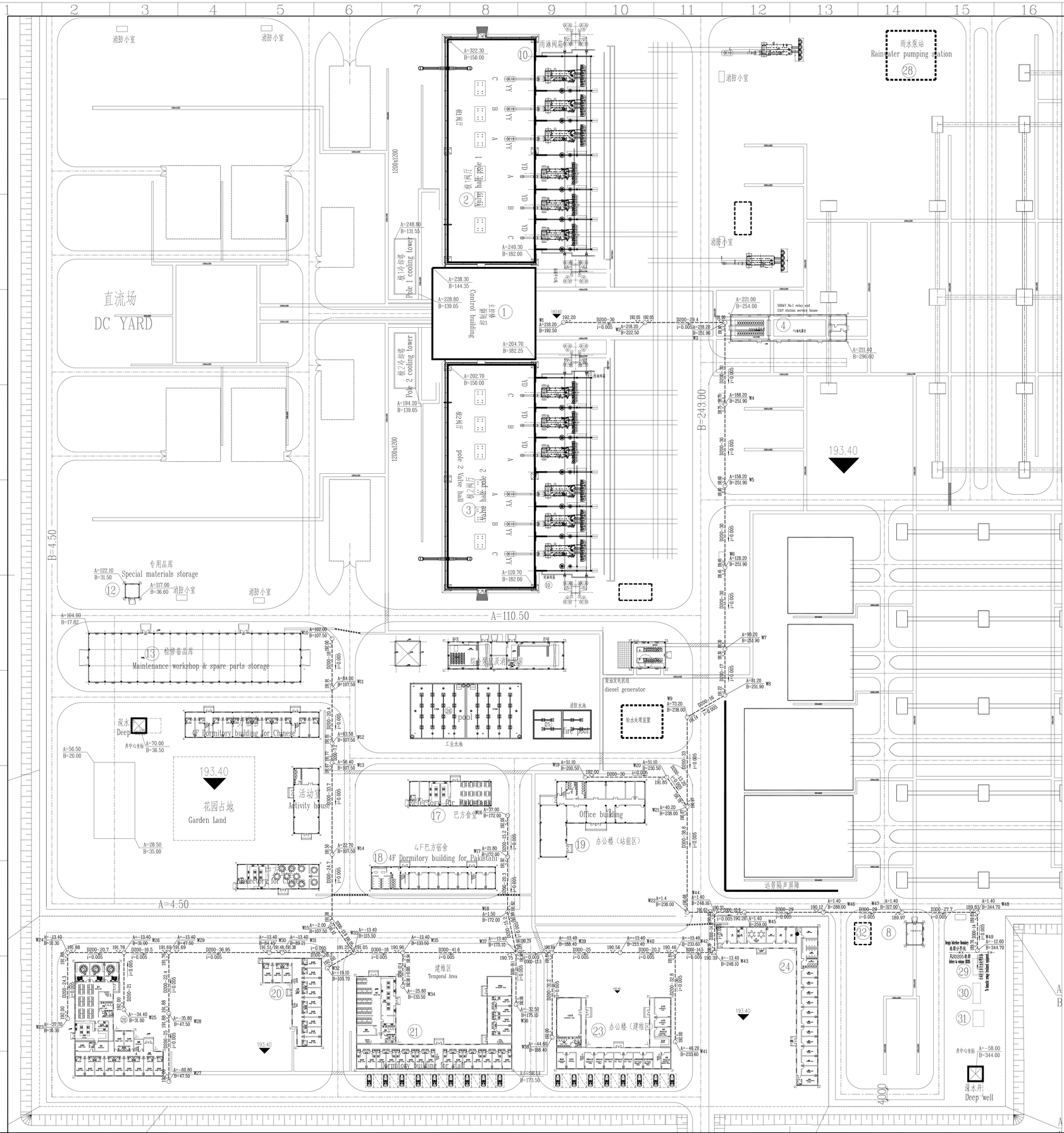
水布 专业 第 02 卷 第 02 册
Hydraulic SPECIALITY VOLUME BOOK

卷册名称 站区生活污水管道布置图
Volume name Outdoor Domestic Sewage Water Piping Layout Drawing

批　准 APPROVED BY	
审　核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	罗
图 纸 3 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图　号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-S0202-001	室外生活污水排水管道平面布置图	A	1		
		Outdoor Domestic Sewage Water Piping Layout Drawing				
2	30-B003701S-S0202-002	室外生活污水检查井汇总表	A	1		
		Summary Table Of Outdoor Domestic Sewage Water Inspection Wells				
3	30-B003701S-S0202-003	室外生活污水排水管道材料表	A	1		
		Outdoor Domestic Sewage Water Piping Material Summary				

注:



说明：

1. 本图卷图标高采用EGM2008高程系统，单位以m计。
2. 本卷图坐标系统采用站区坐标系统，单位以m计；站区坐标系统与巴基斯坦UTM-43N平面坐标系的换算公式详见土建相关卷册。
3. 站内道路为厂区道路，本图仅道路、建筑物、构筑物相对位置仅示意，以土建相关卷册为准。
4. 本工程污水管道采用钢筋混凝土排管，管径以D表示，所注标高为管内底标高。
5. 钢筋混凝土管采用120°砂石基础，其检查井基础详图04S516 11；施工要求及说明详见给水排水标准图04SS163-7。
回填土厚度按施工规范进行。
6. 施工时请参照相关厂家的管道施工安装说明，并符合国家市政排水管道施工规范和各附要求，同时符合GB 50268-2008《给水排水管道工程施工及验收规范》。
7. 本图图中W1为站区污水检查井编号，所有排水管道间检查井的标高确定，严禁倒灌，其它标注方式详见图例。
8. 位于硬化地坪上的污水检查井，井深标高与地坪标高齐平；位于绿地地坪上的污水检查井，井深标高高于所在地坪50mm。
9. 道路地面荷载按汽-15级（加重车齐列）设计，，钢筋混凝土管道按，外压荷载，内水压力满足GB/T11836-2009 II级管。

管径 D	最小管壁厚 mm	裂缝荷载 KN/m	破坏荷载 KN/m	内水压力 MPa
D200	30	15	23	0.10
D300	30	19	29	0.10

10. 如果部分地段的施工荷载过大，施工中仍可能压坏管，井，建施工单位在施工过程中压管多采取保护措施，在施工后（平）检查污水管，井是否完好，若有损坏及沉降管。
11. 所有污水检查井均采用重型铸铁井盖及支座。
12. 施工时请按《给水排水管道工程施工及验收规范》（GB50268-2008）规定对钢筋混凝土排水管进行闭水试验。
检验时，桩件观察，不得有漏水现象。钢筋混凝土管对渗水量应满足Q≤1.25√di ，钢筋混凝土管的允许渗水量应符合以下关系：
（Q：每km管道长度24h的渗水量每m^3；di：管道内径mm）
钢筋混凝土管允许渗水量应符合以下关系：di=200时，渗水量≤17.68m^3/(24h·km)，di=300时，渗水量≤21.65m^3/(24h·km)。
检查井井盖应认真盖好，严防建筑垃圾掉入堵塞下水道。
13. 本工程参照的标准图集：
给水排水标准图集 02SS15 6-7 总说明
给水排水标准图集 02SS15 21 Φ1000mm圆形砖砌污水检查井(盖板式)
给水排水标准图集 97SS01-1 3、4 总说明
给水排水标准图集 97SS01-1 43 Φ700(2Q)重型球墨铸铁支座(A)
给水排水标准图集 97SS01-1 65 球墨铸铁踏步(TQ)
给水排水标准图集 97SS01-1 68 铸铁及塑钢踏步安装图
给水排水标准图集 04SS16 17 D=200~3000钢筋混凝土管180°砂石基础
给水排水标准图集 04SS16 23 D=200~1800钢筋混凝土承插口管 橡胶圈接口
14. 主控楼室内给排水管道布置图详见S0206卷册，其他建筑物室内给排水管道布置图详见相市政图纸。

Notes:

1. EGM2008 elevation is adopted in this volume,unit of elevation is m.
2. The station coordinate(AB) is adopted in this volume,unit of coordinate is m; station coordinate(AB) and UTM-43N system conversion formula refers to Civil Engineering volume.
3. Station road adopts suburban road type.Relative location of roads, buildings and structures in the drawing is only for schematic drawing,details refer to Civil Engineering volume.
4. Material of sewage water piping is RC pipe(Class II), diameters is expressed as symbol D. Pipe elevation is represented with the inner bottom elevation.
5. RC pipe uses 180° sand & gravel foundation,refers to《Water Supply & Discharge Standard Collective Drawing》04SS16 11; Construction requirements and descriptions refer to《Water Supply & Discharge Standard Collective Drawing》04SS163-7. Requirements of oil backfill shall comply with construction standards.
6. Please refer to construction and installation descriptions of relevant manufacturers,comply with relevant standards & rules of national municipal drainage pipeline and《Code for construction and acceptance of water and sewerage pipeline works》(GB50268-2008).
7. Inspection well No. is expressed as W;all drain slope direction is determined according to the start & end point elevation, adverse slope is strictly prohibited,other dimension style refer to graphic symbol.
8. Sewage inspection well that lies on hardened ground has a same top elevation with ground,top elevation of other well that lies on site ground is 50 mm higher than gound elevation.
9. Road ground load level is designed according to vehicle-15 taggravating double car),RC pipe specification, external pressure load, the internal pressure shall meet requirements of GB/T11836-2009 class II.

Diameter D	Minimum Pipe Wall thickness mm	Crack load KN/m	Failure load KN/m	Internal Water Pressure MPa
D200	30	15	23	0.10
D300	30	19	29	0.10

10. If construction load of local site is too large,pipe and well may be crushed,protective measure should be taken as much as possible during construction process. Sewage pipe should be checked afterduringconstruction,any damage should be repaired in time.
11. All sewage inspection well adopt heavy cast iron covers and support.
12. Hydrostatic test for RC drainage pipe shall be completed during construction process according to《Code for construction and acceptance of water and sewerage pipeline works》(GB50268-2008).Leakage is not allowed during visual inspection. Water seepage of RC pipe shall meet Q≤1.25√di ,allowable water seepage shall conform to the following relations: (Q:Water seepage capacity (m^3) per km in 24h;di:inner diameter,mm)
allowable water seepage for RC pipe shall conform to the following relations: di=200, water seepage<17.68m^3/(24h·km),di=300, water seepage<21.65m^3/(24h·km).
Covers should be carefully covered up,in order to prevent drains blocking,dumping building rubbish to inspection well is strictly prohibited.
13. Standard schematic handbook used in this project:
Water Supply & Discharge Standard Collective Drawing 02SS15 6-7 General Description
Water Supply & Discharge Standard Collective Drawing 02SS15 12 Φ1000mm Round brick sewage inspection well(Cover plate)
Water Supply & Discharge Standard Collective Drawing 97SS01-1 3、4 General Description
Water Supply & Discharge Standard Collective Drawing 97SS01-1 43 Φ700(2Q)Heavy ductile iron support(A)
Water Supply & Discharge Standard Collective Drawing 97SS01-1 65 Ductile iron stepping(TQ)
Water Supply & Discharge Standard Collective Drawing 97SS01-1 68 Stepping installation drawing of cast iron and plastic steel
Water Supply & Discharge Standard Collective Drawing 04SS16 17 D=200~3000 RC pipe 180° sand and gravel foundation
Water Supply & Discharge Standard Collective Drawing 04SS16 23 D=200~1800 RC pipe with bell and spigot and Rubber ring joint
14. Water Supply & Drainage Pipe Layout drawing refer to volume S0206,layout drawings of other buildings refer to relevant drawings.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦既帕至卫—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 2 0 2 - 0 0 1 4

	姓名 Name	日期 Date
设计 Designed	卢志	
校核 Checked		
审核 Reviewed		
批准 Approved		

室外生活污水排水管道平面布置图
Outdoor Domestic Sewage Water Piping Layout Drawing

比例 Scale: 1:500 | 图号 DWG NO.: 30-B003701S-S0202-001 | 版本 REV.: A

841X1189

检查井编号 No.	检查井结点 Inspection Well Nodes 平面示意图 Plan Sketch	检查井中心座标 Center Coordinate Of Inspection Well A/B m	干管 Main Pipe D1 管径 Diameter mm	干管 D1 标高 Elevation m	干管 D2 管径 Diameter mm	干管 D2 标高 Elevation m	支管 Branch Pipe ø1 ø3 ø5 管径 Diameter mm	支管 ø1 ø3 ø5 标高 Elevation m	支管 ø2 ø4 ø6 管径 Diameter mm	支管 ø2 ø4 ø6 标高 Elevation m	检查井直径 Inner Diameter Of Inspection Well mm	检查井标高 Elevation Of Inspection Well m 井顶 Top	检查井标高 井底 Bottom	检查井深 Depth Of Inspection Well m	参考图号及备注 Reference Drawing No. & Remarks
W1	D1	218.20 / 192.50	200	192.20							1000	193.60	192.20	1.40	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W2	D1 D2	218.20 / 222.50	200	192.05	200	192.05					1000	193.60	192.05	1.55	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W3	D1 D2	218.20 / 251.90	200	191.90	200	191.90					1000	193.45	191.90	1.55	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W4	D1 D2	188.20 / 251.90	200	191.75	200	191.75					1000	193.45	191.75	1.70	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W5	D1 D2	158.20 / 251.90	200	191.60	200	191.60					1000	193.45	191.60	1.85	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W6	D1 D2	128.20 / 251.90	200	191.45	200	191.45					1000	193.45	191.45	2.00	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W7	D1 D2	98.20 / 251.90	200	191.30	200	191.30					1000	193.45	191.30	2.15	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W8	D1 D2	81.20 / 251.90	200	191.22	200	191.22					1000	193.45	191.22	2.23	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W9	D1 D2	73.20 / 238.00	200	191.14	200	191.14					1000	193.45	191.14	2.31	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W10	D1	102.00 / 107.50	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W11	D1 D2	84.00 / 107.50	200	191.91	200	191.91					1000	193.45	191.91	1.54	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W12	D1 D2	63.58 / 107.50	200	191.81	200	191.81					1000	193.45	191.81	1.64	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W13	D1 D2	56.40 / 107.50	200	191.77	300	191.67					1000	193.45	191.67	1.78	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W14	D1 D2	22.70 / 107.50	300	191.50	300	191.50					1000	193.45	191.50	1.95	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W15	D1 ø2	1.50 / 107.50	300	191.40	300	191.40					1000	193.45	191.40	2.05	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W16	D1	37.00 / 172.00	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W17	D1 D2	21.80 / 172.00	200	191.92	200	191.92					1000	193.45	191.92	1.53	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W18	D1 D2	1.50 / 172.00	200	191.82	200	191.82					1000	193.45	191.82	1.63	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W19	D1	51.10 / 200.50	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W20	D1 ø2	51.10 / 230.50	200	191.85	200	191.85					1000	193.45	191.85	1.60	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W21	ø3 D1 D2	40.20 / 238.00	200	190.97	300	190.87	200	191.78			1000	193.45	190.87	2.58	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W22	D1 D2	10.20 / 251.90	300	190.68	300	190.68					1000	193.45	190.68	2.77	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W23	D1	-37.70 / 10.30	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W24	D2 D1	-13.40 / 10.30	200	191.88	200	191.88					1000	193.45	191.88	1.57	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W25	D1	-34.40 / 31.00	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W26	D1 D2 ø1	-13.40 / 31.00	200	191.78	200	191.78	200	191.90			1000	193.45	191.78	1.67	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W27	D1	-60.80 / 47.50	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W28	D2 D1	-35.80 / 47.50	200	191.88	200	191.88					1000	193.45	191.88	1.57	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W29	D1 D2 ø1	-13.40 / 47.50	200	191.69	200	191.69	200	191.76			1000	193.45	191.69	1.76	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21

检查井编号 No.	检查井结点 Inspection Well Nodes 平面示意图 Plan Sketch	检查井中心座标 Center Coordinate Of Inspection Well A/B m	干管 Main Pipe D1 管径 Diameter mm	干管 D1 标高 Elevation m	干管 D2 管径 Diameter mm	干管 D2 标高 Elevation m	支管 Branch Pipe ø1 ø3 ø5 管径 Diameter mm	支管 ø1 ø3 ø5 标高 Elevation m	支管 ø2 ø4 ø6 管径 Diameter mm	支管 ø2 ø4 ø6 标高 Elevation m	检查井直径 Inner Diameter Of Inspection Well mm	检查井标高 Elevation Of Inspection Well m 井顶 Top	检查井标高 井底 Bottom	检查井深 Depth Of Inspection Well m	参考图号及备注 Reference Drawing No. & Remarks
W30	D1 D2	-13.40 / 84.45	200	191.51	200	191.51					1000	193.45	191.51	1.94	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W31	D1 D2	-13.40 / 89.21	200	191.48	300	191.38					1000	193.45	191.38	2.07	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W32	D1	-19.10 / 105.70	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W33	ø3 D1 D2 ø1	-13.40 / 115.50	300	191.25	300	191.05	200	191.94	300	191.05	1000	193.45	191.05	2.40	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W34	D1	-25.80 / 133.50	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W35	D1 D2 ø1	-13.40 / 133.50	300	190.96	300	190.96	200	191.94			1000	193.45	190.96	2.49	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W36	D1	-32.50 / 175.10	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W37	D1 ø3 D2 ø1	-13.40 / 175.10	300	190.75	300	190.75	200	191.90	200	190.90	1000	193.45	190.75	2.70	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W38	D1	-44.60 / 188.40	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W39	D1 D2 ø1	-13.40 / 188.40	300	190.69	300	190.69	200	191.84			1000	193.45	190.69	2.76	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W40	D1 D2	-13.40 / 213.40	300	190.56	300	190.56					1000	193.45	190.56	2.89	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W41	D1	-46.20 / 233.60	200	192.00							1000	193.45	192.00	1.45	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W42	D1 D2 ø1	-13.40 / 233.60	300	190.46	300	190.46	200	191.84			1000	193.45	190.46	2.99	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W43	D1 D2	-13.40 / 248.10	300	190.39	300	190.39					1000	193.45	190.39	3.06	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W44	ø1 D2 D1	1.40 / 248.10	300	190.32	300	190.32	300	190.63			1000	193.45	190.32	3.13	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W45	ø1 D1 D2	1.40 / 259.00	300	190.26	300	190.26	300	191.70			1000	193.45	190.26	3.19	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W46	D1 D2	1.40 / 288.00	300	190.12	300	190.12					1000	193.45	190.12	3.33	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W47	D1 D2	1.40 / 317.00	300	189.97	300	189.97					1000	193.45	189.97	3.48	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W48	D1 D2	1.40 / 344.70	300	189.83	300	189.83					1000	193.45	189.83	3.62	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21
W49	D1 D2	-12.60 / 344.70	300	189.76	300	189.76					1000	193.45	189.76	3.69	给水排水标准图集 Water Supply & Discharge Standard Collective Drawing 02S515 21

说明：

1. 本图卷册标高采用EGM2008高程系统，单位以m计。
2. 本卷册座标系统采用站区座标系统，单位以m计；站区座标系统与巴基斯坦UTM-43N平面坐标系的换算公式详见土建相关卷册。
3. 位于硬化地坪上的污水检查井，井顶标高与地坪标高齐平；位于场地地坪上的污水检查井，井顶标高高于所在地坪50mm。
4. 其它说明见：B003701S-S0202-01图。

Notes:

1. EGM2008 elevation is adopted in this volume,unit of elevation is m.
2. 2. The station coordinate(AB) is adopted in this volume,unit of coordinate is m; station coordinate and UTM-43N system conversion formula refers to Civil Engineering volume.
3. Sewage inspection well that lies on hardened ground has a same top elevation with ground, top elevation of other well that lies on site ground is 50 mm higher than gound elevation.
4. Other descriptions refer to volume B003701S-S0202-01.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT GROUP CORPORATION CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - S G - 0 2 0 2 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	甲	
校核 Checked		
审核 Reviewed		
批准 Approved		
比例 Scale		图号 DWG NO. 30-B003701S-S0202-002

室外生活污水检查井汇总表

Summary Table Of Outdoor Domestic Sewage Water Inspection Wells

版本 REV. A

594X841

Notes: Note: Pipe fittings (that are unexpectable or with a diameter ≤ DN50) and anti-corrosion materials are included in the length, do not list separately.
注：不可预计管件、小于等于DN50的管件以及防腐材料均计入管长，不另开列。

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量	材料 Material	单重 Single (重量Weight)	总重 Total (重量Weight)	参考图号及备注 Remark
5	Heavy Cast Iron supportor 重型铸铁盖座	ø700	Set 座	49				Water Supply & Discharge Standard Collective Drawing 给水排水标准图集 97S501-1/43
4	Heavy Cast Iron Manhole Cover 重型铸铁井盖	ø700	Set 座	49				Water Supply & Discharge Standard Collective Drawing 给水排水标准图集 97S501-1/41
3	Sewer inspection well 检查井	ø1000	Set 座	49	Bricks 砖砌	–	–	Water Supply & Discharge Standard Collective Drawing 给水排水标准图集 02S515 14
2	Reinforced concrete sewer pipe 钢筋混凝土管（带胶圈,管枕）	D300(Socket flexible interface) Class II D300(承插柔性接口)II级管	m 米	900	RC 钢筋混凝土	–	–	Water Supply & Discharge Standard Collective Drawing 给水排水标准图集 04S516 38 包括橡胶密封圈、管枕、基础 Including rubber seal ring, pillow, foundation.
1	Reinforced concrete sewer pipe 钢筋混凝土管（带胶圈,管枕）	D200(Socket flexible interface) Class II D200(承插柔性接口)II级管	m 米	700	RC 钢筋混凝土	–	–	Water Supply & Discharge Standard Collective Drawing 给水排水标准图集 04S516 38 包括橡胶密封圈、管枕、基础 Including rubber seal ring, pillow, foundation.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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设计 Designed	甲	
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审核 Reviewed		
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比例 Scale		图号 DWG NO.

室外生活污水排水管道材料表
Outdoor Domestic Sewage Water Piping Material Summary

30-B003701S-S0202-003

版本 REV. A

297X420